

STAVE · Small-sample Tests of ABCD Variants and Estimators

Ideas to run on the barrel

Nathan Herling · UA ATLAS HEP · Oct 2026

What this deck covers

Re-run the barrel baseline first, then test four ideas against it: closure terms, a kernel classifier, endcap pretraining, and a kernel model of the joint background.

Context

3 · Four barrel problems

4 · What 2,280 events can show

5 · What ATLAS does

Baseline first

6 · Re-run vanilla BCE and the λ sweep

7 · Trainer hazards

8 · Diagnostics

Ideas

9–12 · Closure terms: the queen, DNC_FLR, the FLR baseline, the λ × μ grid

13–14 · Kernel classifier and its test

15 · Endcap pretraining

16 · Kernels for the joint background

Plan

17 · Order of work

18 · Open decisions

Four barrel problems, not one

A · Unstable fits

Fold KS 0.227 / 0.352 against a null median of 0.079. About 10 optimizer steps per epoch, so epoch-based rules starve the barrel.

B · Weak closure test

2,280 background events, about 456 per fold. Per-fold grading reads 0.10 to 0.15 whatever the model is.

C · The queen is unproven here

One run per μ. On her trained grid she reads worse, not better. The hinge may be shut at about 166 background per batch.

D · Decorrelation costs AUC

DisCo at λ = 10 costs 0.018 on NN1 and 0.033 on NN2. The queen adds another 0.02 to 0.04.

Each lever in this deck is judged by which of A to D it touches.

What 2,280 events can show

BARREL, λ = 10: PASSES

0.081

|nc|med, with 97% of cells inside 2σ

ENDCAP, λ = 5: FAILS

0.048

|nc|med, with 61% of cells inside 2σ

- The barrel passes with the larger error because its error bars are wider.
- σ_pred / A_pred is about 0.08 in the barrel, so roughly 8% per cell is the detection floor.
- No classifier change can show a closure gain smaller than that floor.

Source: Aug 6 DisCo λ sweep, 6×6 background-quantile grid; σ ratio from the Sep barrel pipeline run.

What ATLAS does: split, cut, assign a systematic

Run-2 one-vertex search	Barrel	Endcaps
NN1, NN2 cuts	> 0.5, 0.5	> 0.8, 0.8
Region A observed	245	31
Region A predicted	241 ± 18 ± 6	46 ± 2 ± 9
Region D count	1,176	13,190
ABCD non-closure systematic	2.5%	19.1%

- Separate 128-128 MLPs per region, inputs decorrelated by hand, no DisCo term.
- Vertices at $0.7 < |\eta| < 1.3$ are excluded, not modeled.
- No paper found that transfers information from the high-statistics region to the low one.

arXiv:2503.20445, muon-RoI channel, read in full. The gap claim rests on a limited search.

First: re-run vanilla BCE and the dCor λ sweep

λ	dCor	|nc|med	in 2σ
0 (BCE)	0.333	2.782	0%
1	0.113	0.330	6%
2	0.078	0.247	31%
3	0.077	0.163	69%
5	0.048	0.111	94%
10	0.044	0.081	97%

Why this is not yet a baseline

- One run per λ, seed 7. Two runs at λ = 10, μ = 0 read 0.081 and 0.057
- Graded on the trained grid only
- Trainer hazards still in place
- The Sep toolchain disagreed: it put the barrel at λ = 2 and called λ = 10 dominated

Re-run (proposed): λ in {0, 1, 2, 3, 5, 10, 20}, about 10 seeds each, three closure bands with a shuffle null, both AUCs, worst-mass AUC, dCor, fold KS. Every later idea is graded against this.

Table: Aug 6 barrel sweep, single run per λ, 6×6 background-quantile grid. Config stays quantile + balanced + in-fold scaling.

Trainer hazards to fix before any comparison

Checkpoint on graded rows	The checkpoint is chosen on the same held-out fold that is then graded.
Whole-fold validation loss	Her z^2 grows with event count, so under $\mu > 0$ the checkpoint is selected for closure on the graded rows. Share of her gain explained: not measured.
Unseeded validation cuts	Validation draws her random cuts unseeded, so checkpoint selection is noisy.
Epoch-based budgets	Barrel gets 9 to 11 steps per epoch against 79 to 88 in the endcap. Use step-matched patience (500 max epochs, patience 70, warmup 45).

The first three apply to both regions and bear on the endcap result too.

Diagnostics to run with the baseline

1 · Hinge activity on training batches

Logged on validation only so far: open on 23 to 29% of barrel cuts, 41 to 45% of endcap cuts. Log it on training batches, where she acts.

2 · Shuffle-null closure

Per fold and pooled, in all three bands. Puts a measured number on the barrel detection floor.

3 · Tail counts

Barrel background above q 0.8, 0.9, 0.95 in both scores. Decides whether a tail queen or tail HSIC can say anything.

4 · Sample hygiene

Drop the all-zero MS1Vtx_l1hcal. Resolve 3,506 vs 3,405 rows, the 147 rows with eventWeight ≤ 0, and the four absent preselection columns.

None of these needs new training beyond one instrumented run per region.

Idea: the queen (DNCn_QN) in the barrel

μ	trained grid	edge	tail	NN1 / NN2 AUC
0	0.057	0.304	1.277	0.842 / 0.898
0.5	0.060	0.140	0.875	0.824 / 0.887
1	0.102	0.129	0.168	0.806 / 0.873
2	0.127	0.053	0.391	0.804 / 0.874
5	0.095	0.085	0.624	0.817 / 0.881
10	0.316	0.186	0.062	0.803 / 0.878

- Trained grid: worse with her. Edge and tail: better with her. Neither is established at n = 1.
- Barrel seed sd of |nc|med is about 0.1, larger than most differences in this table.
- This code uses soft counts, so her gradient is live. The dead-gradient bug found Sep 5 was in the fg rebuild.

|nc|med by band, FORGE musweep_barrel_hinged, λ = 10. Edge = one axis at q 0.9/0.95; tail = both.

Idea: DNC_FLR, the noise-floor-subtracted closure term

The term

$$g = (N_B \cdot N_C - N_A \cdot N_D) \,/\, (N_B \cdot N_C + N_A \cdot N_D)$$

$$\text{floor} = \Sigma_R \, (\partial g / \partial N_R)^2 \cdot V_R$$

$$L = g^2 - \text{floor}$$

Soft counts $N_R = \Sigma w$ with Poisson-analog variances $V_R = \Sigma w^2$, measured per batch. Hard weights recover Var = N.

Why it suits the barrel

- No hinge: it gives a gradient at any count and goes to zero smoothly at the floor
- July, 524 barrel background: it stopped at g^2 of 1.6 to 1.8 × floor, while TEC descended to 0.35 × its own floor
- Endcap: converged to TEC, same AUC
- Sep 5: rebuilt in the fg toolchain on soft counts, floor detached; passed the gradient audit

No replicated barrel result for it on the 2,280-event sample is on record. Run it on the same grid as the queen.

Own construction, experimental. Scheme B (floor detached through an EMA) was retained in T6; scheme A retired.

What to re-run for a baseline that includes DNC_FLR

In the repo now (Aug 6)

- Barrel λ sweep, DisCo only: λ 0, 1, 2, 3, 5, 10
- Barrel μ sweeps at λ = 10: queen unhinged and hinged
- All seed 7, one run per cell, fixed 60 epochs
- No barrel factorial, no FLR, no TEC

Add to the code

- DNC_FLR in losses.py: g^2 − floor, floor detached
- A DisCo + FLR rung
- Step-matched early stopping, separate checkpoint split
- Hinge activity logged on training batches

Run on the barrel

- Vanilla BCE
- DisCo λ sweep
- DisCo + queen, λ × μ
- DisCo + FLR, λ × its own weight grid
- About 10 seeds per cell, three closure bands

Units caveat: in this code g is in events ($N_A - N_B \cdot N_C / N_D$), so FLR's weight is not on the queen's scale. Its grid comes from a short scale scan first: [FLR weight levels].

Source: INFO_698_experiments projects/atlas, src/llp and src/runs. Soft counts are already in place, so both terms have a live gradient.

Test: closure terms on a replicated λ × μ grid

Design (proposed)

- Two arms: the FORGE queen and DNC_FLR
- λ in {5, 10}, μ in {0, 1, 2, 5}
- About 10 seeds per cell
- Hazard fixes and step-matched patience applied
- One held-out-cut arm: train q 0.3 to 0.6, grade above

Why 10 seeds

- Seed sd of |nc|med is about 0.1
- At 3 seeds, a cell difference needs 0.16 to reach 2 se
- At 10 seeds, 0.09
- The endcap used 3 with a pooled sd of 0.012

Report per cell: |nc|med in three bands with seed sd, frac2sig, dCor, both AUCs, worst-mass AUC, qn_active.

The stacked GPU trainer ran 1,350 barrel models in 70 min; it needs porting to FORGE's network and loss first.

Idea: a kernel classifier in place of the MLP

How it works

- Build the n × n matrix of pairwise similarities
- Solve for one weight per training event: class-weighted logistic loss plus ridge
- Score = sigmoid of weighted similarity to the training events
- Kernel, width and ridge replace the NN hyperparameters

Why it fits the barrel

- About 3,500 events: an exact fit is cheap
- Convex: one answer, no epochs, no early stopping
- dCor is already a kernel statistic, so a kernel DisCo stays in the family
- A GP version gives a posterior on the score

What it costs

- Changes the classifier, not the event count
- Adding a dCor penalty between two of them breaks convexity
- Closure terms draw cuts per batch; a full-sample fit needs a new formulation
- Kernel in the barrel, NN in the endcap is harder to defend in review

Hybrid option: NN trunk as the feature map with a kernel or GP head, close to the FORGE last-layer refit.

Test: kernel against MLP, vanilla rung first

Arms

- Linear, as the floor
- RBF at median distance, half, double
- RBF with per-feature widths
- Matérn 5/2
- 128-128 MLP, as the reference

Protocol

- Class-weighted kernel logistic, scaling in fold
- Width and ridge chosen by inner CV
- Variance over repeated fold partitions, both arms
- Selection and reporting seeds disjoint

Win and stop (proposed)

- Win: both AUCs and worst-mass AUC within 1 pooled sd of the MLP, and partition sd of |nc|med cut by [factor]
- Stop: no kernel meets that

A convex fit has zero seed variance by construction, so seed spread alone is not a fair score.

If linear is within noise of the MLP, the kernel sweep is moot at this sample size.

Idea: pretrain on the endcap, apply to the barrel

Four variants

- 0 · Endcap only: train on endcap, score the barrel with that network unchanged
- 1 · Fine-tune: train on endcap, continue on barrel
- 2 · Frozen trunk: endcap trunk fixed, new barrel head
- 3 · Shared trunk: one trunk on both, a head per region

Why try it

- Endcap has 24,850 background events against the barrel's 2,280
- Variant 0 shows what carries over before any barrel data is used
- No ATLAS or CMS paper found doing this across detector regions

What stands in the way

- Registries differ: 30 barrel features, 26 endcap. Only the shared [n] transfer
- Vertexing differs: curvature fits in the barrel, straight lines in the endcap
- It adds training signal, not barrel events for the closure test

Not run this summer: every barrel run in the repo trained on barrel data. Control: a from-scratch barrel MLP on the same features.

arXiv:2412.10665, arXiv:2605.07471: read from search excerpts only. The gap claim rests on a limited search.

Idea: kernels for the joint background

Where it starts

- MaxEnt mk1: per-feature marginals, then a joint by product, Gaussian copula or GMM copula
- Samples go through the frozen NN1 and NN2 to rebuild the quadrants
- ABCD assumes the product; the joint is where non-closure lives

The kernel version

- Model the dependence between the NN1 and NN2 feature blocks with kernel features as the MaxEnt constraints
- Or model it directly in the 2-D score plane, which 2,280 events can support
- Model B, C and D; region A is the result

What to check first

- Unbinned HSIC permutation test against the 6×6 grid: does it have more power at this sample size?
- Tail-weighted against unweighted, kernel held fixed
- Barrel background above q 0.9 in both scores: enough to fit a tail?

This idea changes the estimate, not the classifier. It runs on frozen scores from the baseline re-run.

Idea stage: a test plan comes before any build. A kernel swap inside dCor is not this idea; the corner is about 4% of pairs under any kernel.

Order of work

0 · Baseline re-run	Vanilla BCE and the DisCo λ sweep, replicated, with the trainer fixes and diagnostics. Everything else is graded against it.
1 · Closure terms	The queen and DNC_FLR on the same λ × μ grid.
2 · Kernel classifier	Vanilla rung against the MLP. Needs only step 0, so it runs beside step 1.
3 · Endcap pretraining	Vanilla rung against a from-scratch barrel MLP. Also runs beside step 1.
4 · Kernel joint background	Test plan first. Uses frozen scores from step 0, so it does not wait on 1 to 3.

Open decisions

1. Baseline λ grid: stop at 10 or extend to 20? Seeds per cell?
2. Checkpoint selection: a third split, or fixed-step training with no selection?
3. Which toolchain is the baseline built on: the FORGE trainer or the Sep fg package?
4. DNC_FLR: scheme B only, and at which μ levels?
5. Pretraining: which variant first, and which features count as shared?
6. Joint background: feature space or score plane first?
7. A win criterion per idea, written down before its run.

Sources: Barrel/Endcap literature pass (Oct 5), QUEEN_BARREL_HANDOFF_mk1, Jul DNC family notes, Aug 6-7 sweeps, Sep barrel runs.